

**Do prediction markets out forecast the
National weather service? evidence from
Kalshi's prediction markets.**

Victor Singh

August 27, 2026

Contents

1	Introduction	1
2	Literature Review	1
3	Data	2
4	Methodology	3
5	Results	6
5.1	Why the market wins	6
5.2	Whole-distribution and degree-scale results	7
5.3	Mechanism: who moves first	8
6	Robustness Checks	9
7	Discussion & limitations	9
8	Conclusion	11
	REFERENCES	13

List of Tables

1	Sample summary statistics	3
2	Forecast skill by horizon	6
3	Brier score decomposition	6
4	Calibration regressions	8
5	Point-forecast mean absolute error	8

1 Introduction

As much as they walk, talk, and act like gambling platforms, Kalshi and Polymarket are actually classified by the US government as derivatives trading venues, the same designation as commodity futures markets. But where futures trading was long reserved for the most audacious of Wall Street brokers, the culturalization of prediction markets means any average American can now bet on almost anything, from what a politician says in a speech to when the US military next bombs a foreign country. And while these markets present obvious risks of overindulgence, their enthusiasts posit a more flattering possibility, that market odds capture objective chance better than experts do. The logic is that any individual bettor has some hint to a true probability of an event mixed with some idiosyncratic bias, but aggregate a thousand bettors and the biases erode, leaving prices that reflect genuine probabilities. The favorite exhibit is the 2024 election, when much of the polling and pundit class leaned toward a Harris victory while the betting markets had Trump as a clear favorite; a case where a broad survey of people with money at stake outperformed the insiders. But a single dramatic call cannot distinguish a superior forecasting mechanism from a lucky strike. Elections are rare events; testing the reputation properly requires a domain where the market makes hundreds of verifiable predictions against a serious institutional rival.

In 1984, Richard Roll investigated whether movements in orange juice futures anticipated National Weather Service (NWS) forecast errors for central Florida, where nearly all domestic orange production was concentrated. His theory was that information the NWS had not yet incorporated was being traded into the commodity price. The signal was necessarily indirect, filtered through the total futures price, raising endogeneity concerns. A modern revisit is now possible through prediction markets, which allow for a direct channel. The contract's payoff is the explicit temperature event, so the market price can be read as a stated probability of the weather itself. This paper asks whether that decentralized price mechanism aggregates and acts on information faster and more accurately than the institutional alternative, the National Weather Service's own forecast model.

2 Literature Review

This paper sits at the intersection of two questions: whether market prices embed forecasting information, and how, probabilistic forecasts should be judged. The founding result on the first question is Roll (1984), who showed that returns on frozen orange juice futures predicted subsequent errors and revisions in National Weather Service temperature forecasts

for central Florida, traders were incorporating weather information the official forecast had not yet absorbed. But Roll’s evidence was necessarily indirect: with no market on the weather itself, temperature beliefs had to be inferred from a commodity price entangled with other factors like supply, demand, and inventory conditions, and the market’s weather forecast could never be stated, scored, or checked for calibration. The broader literature that followed found prediction markets generally rivaling or beating polls and professional forecasters (Snowberg, Wolfers and Zitzewitz, 2013), with miscalibration appearing mainly at long horizons (Page and Clemen, 2013) and the choice of aggregation mechanism mattering less than the aggregation itself (Atanasov et al., 2017). Most directly related, Burgi, Deng and Whelan (2026) provided the first systematic study of Kalshi, documenting informative prices but a persistent longshot bias across most Kalshi market contract categories, weather included. My results provide a sharp counterpoint: applying their own calibration test to temperature markets, I find the apparent bias is an artifact of the Kalshi market structure, something I will explain later in the paper. The verification framework itself comes from the meteorological literature: following Murphy (1993) and Murphy and Winkler (1987), forecast quality is treated into multiple parts and decomposed Brier scores are broken down into reliability and resolution, to ultimately make predictions about calibration are more discrimination; completing, four decades later, the direct test Roll’s data could not support.

3 Data

Market data comes from Kalshi’s daily high-temperature series for New York City (KX-HIGHNY). On each day, Kalshi hosts a market that consists of six mutually exclusive options covering the fields of possible high temperatures, two degree bins flank the official forecast with open ended tails on either side (e.g., “84–85°F,” “86°F or above”). Each open option trades as a binary contract, if you enter for the listed price your payoff is either 1 or 0, so the contract options constitute a probability distribution over the day’s possible outcomes. Through Kalshi’s public API data is collected via settled contracts and their hourly price updates, the final sample spans 135 event-days across 21 weeks in 2025-2026, a total of 810 contracts with each day having a complete six temperature bins. Summation of total bin prices have an approximately normal distribution, with a median of 1.02. Each observation is a contract–snapshot pair, with snapshots taken 24, 12, and 4 hours before the time of the daily maximum (approximately 3 p.m. local), plus one post-peak snapshot 6 hours after it. Of 3,192 candidate observations, two pre-registered filters are used to refine observations. Contracts are discarded when the most recent bet is more than three hours

old, or if contracts have a total volume below \$1,000. The final analysis sample contains 3,214 contract–snapshot observations.

Kalshi’s institutional rival is the National Blend of Models (NBS), the National Weather Service’s predictive model, obtained from the Iowa Environmental Mesonet archive for the Central Park station (KNYC). Each run is timestamped at issuance, enabling the matched-information design described below: at every snapshot, the market is compared only against the most recent NBS run issued by that moment, so neither side is graded on information the other could not have had. Runs whose coverage of the target date begins after 10 a.m. local are excluded. Because NBS issues a point forecast rather than probabilities, each forecast is converted into bin probabilities by fitting a normal distribution over the bias-corrected point, with spread and bias estimated from the model’s historical errors at each lead time (archived runs from July 2023 onward, always excluding the year being scored). Daily observed highs from 2005 onward, from the same archive, verify settlement and construct the climatological baseline. Each bin’s historical frequency among days within a week of the same calendar date, excluding the target year.

Table 1: Sample summary statistics

	Value
Event-days	137
Contracts (bins)	822
Calendar span	Apr 2026–Aug 2026 (21 ISO weeks)
Contract–snapshot observations	3,214
Day ladder volume (\$), median [IQR]	218,393 [166,861–273,761]
Ladder mass, median [5%–95%]	1.02 [0.98–1.07]
Share of prices at grid edge	0.42
Qualifying NBM runs	15,217
Observed daily highs (from 2005)	7,900

4 Methodology

The unit of observation is a contract-time stamp. For each event in a day, a snapshot of the current market structure several times relative to the time of the daily high are taken, 24, 12 and 4 hours before the peak, plus one snapshot after the daily high resolves (~6 hours after). At each recorded time stamp, three separate forecasters state a probability for each temperature bin. The prediction market’s assumed probability is its contract price directly, a bin contract listed at 30 cents is a 30 percent claim. The National Weather Services model,

National Blend of Models (NBM) shows a point prediction, the point prediction is taken at each time snapshot (24,12,4 hours out) intentionally at the same time as each market snapshot to satisfy the assumption of matched information. As the NBM does not issue probabilities a bell curve is constructed through the models own track record, from three years of archived forecasts not using overlapping data from the comparison days. This is done through measuring how far off forecasts at each time stamp normally land, then construct a bell curve with that calculated spread over the forecast under a Gaussian error model. The probability for each bin is the share of the curve falling inside each bin. Lastly I add a baseline, Climatology, which uses the historical frequency of each temperature bin among historic data within a week of the same date in the tested year.

$$p_k^{\text{NBS}} = \Phi\left(\frac{h_k + 0.5 - (F - \hat{b}_\ell)}{\hat{\sigma}_\ell}\right) - \Phi\left(\frac{l_k - 0.5 - (F - \hat{b}_\ell)}{\hat{\sigma}_\ell}\right) \quad (1)$$

As previously stated as the NWS issues a point prediction and not a probabilistic temperature distribution, to remedy this I construct temperature bin probabilities through a Gaussian error model. For a bin with integer bounds $(l_k, h_k]$ and a forecast whose lead time falls in bucket (buckets are a time before the high of a day in which the data is collect bucket $(0, 10]$, $(10, 28]$...) ℓ , where Φ is the standard normal CDF and the bias and spread are the mean and standard deviation of archived forecast errors in that lead bucket. Both parameters are estimated using only errors from other years. For open-ended tail bins the corresponding Φ term is set to 0 or 1, and the half-degree offsets are used to reflect capture integer settlement at each bin, $(l_k, h_k]$ owns the interval $(l_k - 0.5, h_k + 0.5]$, so adjacent bins are created along the mean point prediction and the probabilities sum to one exactly.

Forecasts are judged via Brier scores.

$$BS = \frac{1}{N} \sum_i (p_i - y_i)^2 \quad (2)$$

Where $Y \in \{0, 1\}$ is dependent on the contract's outcome. Brier scores punish far off predictions more strongly than near misses, and it is this squared penalty that makes the score strictly proper; the expected loss is minimized only by reporting one's true belief (Gneiting and Raftery, 2007). Performance is ranked by the Brier skill score, $BSS = 1 - BS_{\text{mkt}}/BS_{\text{ref}}$, where first brier scores are calculated for each model and then entered into the above equation. Positive values from the BSS indicate the market outperformed the reference model, while negative scores indicate the reference model's superior accuracy.

Additionally, Brier scores are decomposed following Murphy (1973) breaking the score

into reliability, resolution and uncertainty. Reliability measures how close the predicted probabilities are to the actual relative frequencies in each subgroup. Resolution incorporates the forecaster’s ability to sort events into subsets with different relative frequencies compared to the overall sample mean. Uncertainty measures incorporate the variance of observations defined as $\bar{y}(1 - \bar{y})$, which depends only on the base rate of the event. A calibration test is used with the regression $y_i - P_i = \alpha + \psi P_i$, on yes-side contracts testing whether the price-probability for any market truly wins the expected amount of time (contract at .30 cents wins 30% of the time). I test whether a contract’s gap between its outcome and its price varies with the price level. Under an honest report of probability/ price the slope should be 0. Because of Kashi’s market structures, bets cannot be taken with a below 1% probability ($P < 1$ cent), so I repeat this regression while excluding prices within 1 and then 3 cents of $[0, 100]$. Echoing Roll (1984), I test whether market prices anticipate the NWS model updates. Between any two NWS updates the model cannot change its prediction while the market keeps trading. For each interval I regress the model’s revision ΔF , on the change in the market’s implied mean temperature (ΔM).

$$\Delta F = \beta_0 + \beta_1 \Delta M + u \tag{3}$$

A positive slope would show that market movements during the NWS update gap predicts change of the next forecast update.

The six bins of an event-day are mutually exclusive events on a single outcome, and because consecutive days share weather patterns, the data contain few truly independent observations. All uncertainty estimates therefore come from resampling entire days at random (and, for robustness, entire weeks), rather than individual rows. Concretely, the 137 event-days are drawn with replacement 2,000 times, so the dependence within a day is preserved in every resample. Each resample produces one alternate estimate of the statistic of interest, and the middle 95 percent of the resulting distribution forms the reported confidence interval. Additionally, If results show markets indicate some predictive advantage, as an economic complement I evaluate potential strategies that trades whenever the NBM probability and the market price diverge by more than 5 cents after Kalshi’s platform fee of $0.07 P(1 - P)$ per contract, under the a potential finding that the market is a better forecaster, this strategy returns whether the models information retains any exploitable value at market prices.

Table 2: Forecast skill by horizon

Horizon	N	Brier			BSS vs NBM			BSS vs
		Market	NBM	Clim	Est.	95% CI (day)	95% CI (week)	Clim
T-24h	774	0.112	0.132	0.175	0.149	[0.097, 0.201]	[0.082, 0.207]	0.357
T-12h	774	0.102	0.126	0.175	0.194	[0.126, 0.265]	[0.103, 0.274]	0.418
T-4h	774	0.087	0.127	0.175	0.313	[0.230, 0.394]	[0.189, 0.415]	0.503
T+6h (post-peak)	774	0.000	0.127	0.175	1.000	[0.999, 1.000]	[0.999, 1.000]	1.000

5 Results

Brier scores are reported at three pre-peak snapshots on a panel of 129 event-days observed at every time horizon. Comparing the market’s Brier skill score against the NWS model reveals a consistent market advantage. At 24 hours before the daily high, the horizon at which both forecasts are competing on the same information, the market achieves a BSS of +0.149 [0.096, 0.201], estimates are not an artifact of correlated days being treated as independent as the BSS of week-level blocking [0.083, 0.207] still archives statistical significance. By 12 hours out the market advantage widens to +0.194 [0.126, 0.264]; both forecasters do achieve an increase in Brier Score as the overnight information provides higher clarity. At 4 hours before the relative daily peak the Brier score for markets jumped to 0.0869, this Brier score cannot be used to calculate a BSS as the NWS stops releasing new daily higher predictions at 10AM, comparing them both would violate the equal information assumption.

5.1 Why the market wins

Decomposing the Brier scores gives a good idea to where the advantage of the market lies. At the T-24 hour window the market achieves a lower reliability penalty (0.0014 vs NWS model 0.0036), meaning the market’s stated probabilities are closer to the frequencies of actual events. The market also scores a better resolution score at 0.0276 against the NWS model’s 0.0101, a higher score is better in this instance as the market can take information and predict non-trivial changes in temperature better than the NWS model can. With a market one might assume a trade off where it adapts to changes quicker and has a better resolution but at the cost of a lower reliability, but this is clearly not the case and pushes towards the claim that market’s edge lies in superior predictive ability rather than an artifact from its flexibility.

Table 3: Brier score decomposition

Horizon	Forecaster	Brier	Reliability ↓	Resolution ↑	Uncertainty
T-24h	Market	0.1120	0.0014	0.0276	0.1390
	NBM	0.1320	0.0036	0.0101	0.1390
	Climatology	0.1750	0.0378	0.0020	0.1390
T-12h	Market	0.1020	0.0016	0.0376	0.1390
	NBM	0.1260	0.0075	0.0177	0.1390
	Climatology	0.1750	0.0378	0.0020	0.1390

A full-sample calibration test reproduces the classic longshot signature of a negative intercept and a positive slope ($\alpha = -0.0116$, $\psi = +0.0465$ at the day-ahead snapshot), meaning cheaply priced contracts win less often than their prices imply. While some long-shot bias could definitely be in effect, the market betting structure of Kalshi reveals a different explanation. Kalshi provides contracts only in whole cents, leading to mechanically limited and overpriced bins for the tail contracts. Additionally as each event-day partitions the temperature range into six mutually exclusive bins, a large share of observations sit at exactly these boundary tails. I then repeat the regression while excluding prices within one and then three cents of the boundaries. The intercept attenuates toward zero; a joint test shows unbiasedness can no longer be rejected ($\alpha = -0.0082$, $p = 0.78$). It can be claimed that away from the absolute price edges, where prices can reflect probabilities better, Kalshi’s temperature contracts are statistically indistinguishable from event probabilities. This evidence shows that on any given contract long shot bias is relatively low and only present for extreme long-shot cases, within 3 cents of the boundary. Overall the evidence demonstrates that long shot behavior bias is not a major market shaper, consistent with the presence of automated betting algorithms.

5.2 Whole-distribution and degree-scale results

Two further results confirm that the ranking does not depend on how the forecasts are scored. As Brier score evaluates each temperature bin independently, forecasts that miss by one bin are treated equally to those that miss by more. The ranked probability score (RPS) instead grades each day’s entire probability as a single distribution, penalising forecasts in how far their probability estimate is from the recorded temperature. Under this set the market’s advantage is clear, achieving a RPS of +0.365 against the NWS model at the 24 hour snapshot. The point is that the market’s advantage is still present once we distinguish the difference between near misses and distant misses. Results thus are consistent with

Table 4: Calibration regressions

	<i>Dependent variable: resid</i>		
	Full sample	Excl. $\leq .01/\geq .99$	Excl. $\leq .03/\geq .97$
alpha (constant)	−0.012 (0.010)	−0.013 (0.013)	−0.008 (0.020)
psi (price)	0.047 (0.059)	0.052 (0.068)	0.038 (0.090)
Joint p (cluster bootstrap)	0.00	0.00	0.78
Event-day clusters	137	136	135
Observations	822	730	601
R^2	0.001	0.001	0.000
Adjusted R^2	−0.001	−0.001	−0.001
Residual Std. Error	0.327	0.347	0.382
F Statistic	0.626	0.574	0.183

its errors being more tightly concentrated around the realized outcome. As prices across each daily market sum to a median of 1.02, a small overround that requires renormalizing each ladder to unit mass before it can be scored as a probability distribution. Taking each forecaster’s implied mean temperature, the market misses the realized daily high by 1.74°F a day in advance against the model’s 2.46°F only where the forecasters center their prediction.

Table 5: Point-forecast mean absolute error

Horizon	Days	Market (°F)	NBM (°F)	Gap
T-24h	129	1.70	2.41	+0.71
T-12h	129	1.41	2.27	+0.86
T-4h	129	1.15	2.28	+1.13
T+6h (post-peak)	129	0.73	2.28	+1.55

5.3 Mechanism: who moves first

Additionally Roll-style regressions are run to reveal that in the intervals between NWS model updates, price shifts in the market predict the direction of the next update. Across 374 run-pairs spanning 137 event-days, the regression produces a slope of +0.302 [0.075, 0.544]: when the market’s implied temperature mean shifts by 1°F, the next NWS forecast shifts in the same direction by roughly 0.30°F. The magnitude is not the market anticipating 30 percent of the revision, as the implied mean and the model’s point forecast are different objects, so the coefficient should be read as establishing direction rather than sizing the share of the

next revision.

6 Robustness Checks

Robustness tests were run to ensure results are not an artifact of choices in building the tests. I re-run the comparison in three different ways to see whether the market’s advantage depends on any single specification. All three tests are reported using the twelve-hour snapshots.

The first test I run concerns how a price for a contract is chosen. At any moment a contract has both a standing bid and ask, prices on the buy and sell side, and a last traded price, these often disagree slightly. In my main results I chose to use the midpoint of the buy and sell side, but this midpoint can update without any new trades leaving a possible discrepancy. Re-running the Brier Skill Score tests on the last traded price attenuates the finding, moving from +0.195 to +0.189, although the new specification lowers the finding it is still confidently within the markets favor.

The second test concerns how a recent price must be offered to count. If no one has traded a contract for hours, the last recorded price may no longer actually reflect what probability traders believe. My main result discards any quote more than three hours old. Re running that test with both an expanded window of 6 hours and 1 hour leaves the main skill score result unchanged. Which is unsurprising, since fewer of one percent of total bets in the sample are more than an hour old. These markets have adjusted prices and thus bettors change their belief almost continuously before a window closes.

The third test concerns the NWS weather models constructed probabilities using the Gaussian error model. The Gaussian error model assumes that the models forecasts are equally uncertain on every day at a given snapshot time, in reality this is not the case as some days sitting on the borders of seasons most likely have higher variation. So I refit the NWS model separately into each month, which lets the model express more or less confidence in predictions depending on the time of year. The market Skill score advantage falls slightly from 0.195 to 0.183. This fall is not enough to assume that the NWS constructed model’s lack of changeable variation is what produces the market’s advantage.

7 Discussion & limitations

The data used here are limited to contracts, timestamps, and prices as aggregate information; there is nothing about the characteristics of the people taking each side. Two scenarios

would explain the findings. Many individual traders whose biases cancel out into accurate prices, or a handful of automated systems running their own arbitrage betting systems. Temperature is perhaps the most automatable market Kalshi hosts as the inputs are public, the settlement is objective, and the optimal strategy can largely be derived from public data. That considered, there is a real chance that a substantial share of volume is placed algorithmically. The absence of a longshot bias at prices near the corners points the same way. It is worth noting that the two scenarios are not mutually exclusive. Arbitrage still requires something to arbitrage against, so a market dominated by automated participants still presupposes a baseline of human pricing error for those systems to correct. What the data cannot do is separate the two, under the first scenario the result is all about the wisdom of a crowds effect (law of large numbers), under the second it is about a few well-built models.

The Gaussian distribution that I use to create the probabilistic curve for the NWS itself is another place to discuss. Any miscalibration of the NWS probability distribution would degrade the NWS model's score rather than the market's. But the reliability penalty, in which a perfectly calibrated distribution would push to zero, is not large enough to have any major change if attenuated (the model's reliability penalty of 0.0036 against a Brier gap of 0.0196). Although, holding forecast dispersion constant within each lead-time bucket limits how sharply the model can express its confidence intervals. Two additional considerations suggest the effect is limited. One, the market's resolution advantage is roughly threefold. Two, the market's implied mean outperforms the model's point forecast by 0.7°F, a comparison in which distributional shape would not change. The appropriate claim should not be that traders out-forecast the National Weather Service, but that market prices aggregate weather information more effectively and at a better temporal resolution than the statistical algorithms of the NWS.

The horizon results and the revision regression point to a single mechanism behind the market's advantage: the market updates continuously while the model updates on a schedule. This means part of the finding may reflect temporal updating rather than superior predictive ability. Two qualifications are worth stating. At both the 24-hour and 12-hour snapshots the most recent qualifying run was on average only about forty minutes old, so the day-ahead comparison is not one of an outdated rival against a live one. Additionally only a small fraction of market days had contracts trading as far out as 36 hours, so whether the market retains its advantage at longer lead times is untestable. That limitation is informative as prediction markets forecast only where someone has chosen to list a contract, so whatever their possible accuracy, they cannot substitute for an institution that covers every horizon.

The calibration results are, on closer inspection, as much a finding about market design as

about trader behavior. As Kalshi quotes in whole cents, contracts whose true probability is a fraction of one percent cannot be priced below half a cent. Because each event-day partitions the temperature range into six bins, a large share of observations sit at exactly that remote tail boundary. This has two implications. Methodologically, studies documenting longshot bias on tick-constrained platforms may be measuring the artifact of tick-price constrained betting rather than bettor behavior. Practically, a finer tick size on low-probability contracts would let prices in the tails tell the truth, which matters for anyone who takes these markets seriously as probability estimates rather than as entertainment (which is not that big of a concern).

Finally, these findings need replication at a larger scale. This study covers one station over roughly six months, and NYC’s weather is comparatively consistent. In regions with more volatile or less predictable weather it is unclear whether the result would hold. Replication on other weather markets, not just daily highs, should also be tested further. A broader study across many cities, several years, and multiple weather variables is certainly warranted before treating this as a general property of prediction markets.

8 Conclusion

In a game of poker, classic Texas hold’em, a gambler might take any number of potential cues as reliable information. Any tell, sweat, previous hand, or board state provides, in the gambler’s mind, some insight into what their opponents might hold. That same idea prompted this investigation into weather prediction markets: a single bettor might receive a cue that offers slight insight into the true weather for a day, and averaged across a market of a thousand gamblers, those signals could compile into an accurate probability for that day’s outcome. The results back up that theoretical baseline. Kalshi’s daily temperature markets outperform the National Weather Service’s guidance model, beating it on both resolution and reliability, and market prices anticipate the direction of the model’s own subsequent revisions.

Caveats to those findings exist. Available data cannot establish whether the accuracy reflects the true canceling out of bias across individual bettors or findings more closely reflect automated systems exploiting human forecasting error. The rival the market is tested against is constructed by me and does not reflect actual released probability, additionally part of the market’s advantage is its ability to constantly update and change while the NWS rival is on a schedule. Additionally, further investigation is needed to judge whether findings can be extrapolated to a larger population through looking at other weather markets in different

cities, looking at other weather based markets not just temperature and finally expanding to other markets not weather based.

What evidence does establish through is not necessarily trivial. When a market is priced over a well-defined outcome, its prices are informative, predictive of the true outcome, better calibrated and more discriminating than the NWS forecasts. When Roll published his paper 40 years ago he could only infer that weather information was embedded in the prices of orange juice futures, filtered through many possible other explanations; endogeneity remained a persistent issue. The contracts studies here pay out on the temperature clearly, which makes the market forecast able to be read off directly, scored and judged against the NWS. On that scoreboard, over one summer in one city, the gamblers won.

REFERENCES

- Atanasov, Pavel, Phillip Rescober, Eric Stone, Samuel A. Swift, Emile Servan-Schreiber, Philip Tetlock, Lyle Ungar, and Barbara Mellers. 2017. “Distilling the Wisdom of Crowds: Prediction Markets vs. Prediction Polls.” *Management Science* 63 (3): 691–706.
- Brier, Glenn W. 1950. “Verification of Forecasts Expressed in Terms of Probability.” *Monthly Weather Review* 78 (1): 1–3.
- Burgi, Constantin, Zhijie Deng, and Karl Whelan. 2026. “Makers and Takers: The Economics of the Kalshi Prediction Market.” Working paper.
- Glahn, Harry R., and Dale A. Lowry. 1972. “The Use of Model Output Statistics (MOS) in Objective Weather Forecasting.” *Journal of Applied Meteorology* 11 (8): 1203–11.
- Gneiting, Tilmann, and Adrian E. Raftery. 2007. “Strictly Proper Scoring Rules, Prediction, and Estimation.” *Journal of the American Statistical Association* 102 (477): 359–78.
- Gneiting, Tilmann, Adrian E. Raftery, Anton H. Westveld III, and Tom Goldman. 2005. “Calibrated Probabilistic Forecasting Using Ensemble Model Output Statistics and Minimum CRPS Estimation.” *Monthly Weather Review* 133 (5): 1098–1118.
- Liang, Kung-Yee, and Scott L. Zeger. 1986. “Longitudinal Data Analysis Using Generalized Linear Models.” *Biometrika* 73 (1): 13–22.
- Mincer, Jacob, and Victor Zarnowitz. 1969. “The Evaluation of Economic Forecasts.” In *Economic Forecasts and Expectations*, edited by Jacob Mincer, 3–46. New York: National Bureau of Economic Research.
- Murphy, Allan H. 1973. “A New Vector Partition of the Probability Score.” *Journal of Applied Meteorology* 12 (4): 595–600.
- Murphy, Allan H. 1993. “What Is a Good Forecast? An Essay on the Nature of Goodness in Weather Forecasting.” *Weather and Forecasting* 8 (2): 281–93.
- Murphy, Allan H., and Robert L. Winkler. 1987. “A General Framework for Forecast Verification.” *Monthly Weather Review* 115 (7): 1330–38.
- Page, Lionel, and Robert T. Clemen. 2013. “Do Prediction Markets Produce Well-Calibrated Probability Forecasts?” *Economic Journal* 123 (568): 491–513.
- Roll, Richard. 1984. “Orange Juice and Weather.” *American Economic Review* 74 (5): 861–80.

Snowberg, Erik, and Justin Wolfers. 2010. “Explaining the Favorite–Longshot Bias: Is It Risk-Love or Misperceptions?” *Journal of Political Economy* 118 (4): 723–46.

Snowberg, Erik, Justin Wolfers, and Eric Zitzewitz. 2013. “Prediction Markets for Economic Forecasting.” In *Handbook of Economic Forecasting*, vol. 2A, edited by Graham Elliott and Allan Timmermann, 657–87. Amsterdam: Elsevier.

Wolfers, Justin, and Eric Zitzewitz. 2004. “Prediction Markets.” *Journal of Economic Perspectives* 18 (2): 107–26.